INS3143 CYBER ATTACK AND MITIGATION
TEST 2

MULTIPLE-CHOICE QUESTIONS (70 MARKS)

Choose The Best Answer

Topic 1 : Port Scanning

1. **Which tool is most commonly used for port scanning?**
   A. Wireshark
   B. Nmap
   C. Snort
   D. Burp Suite

2. **Which scan is known as a "half-open" scan?**
   A. FIN Scan
   B. SYN Scan
   C. TCP Connect Scan
   D. ACK Scan

3. **Which type of scan is typically used to bypass simple packet filters?**
   A. UDP Scan
   B. Null Scan
   C. TCP Connect Scan
   D. Xmas Scan

4. **Port 443 is used by which protocol?**
   A. HTTP
   B. SSH
   C. HTTPS
   D. FTP

5. **Which scan completes the full three-way handshake?**
   A. SYN Scan
   B. FIN Scan
   C. TCP Connect Scan
   D. Stealth Scan

6. **What does a closed port indicate?**
   A. No service is running
   B. Service is reachable
   C. Firewall is blocking the port
   D. The system is down

7. **Which scan type is the slowest?**
   A. UDP Scan
   B. SYN Scan
   C. FIN Scan
   D. Xmas Scan

8. **Masscan is known for:**
   A. Deep packet inspection
   B. High-speed port scanning
   C. Malware analysis
   D. Packet crafting

9. **Which protocol does UDP scanning target?**
   A. Connection-based services
   B. Connectionless services
   C. Encrypted services
   D. Stateful services

10. **A filtered port means:**
    A. Port is open
    B. Port is closed
    C. Firewall prevents checking
    D. Host is inactive

Topic 2 : Enumeration

1. **Enumeration occurs in which phase of an attack?**
   A. Gaining Access
   B. Reconnaissance
   C. Post-Exploitation
   D. Scanning

2. **Which protocol is used for directory queries?**
   A. SNMP
   B. LDAP
   C. SMTP
   D. SMB

3. **Which tool is used for SMB enumeration?**
   A. Nikto
   B. Enum4linux
   C. Burp Suite
   D. Zenmap

4. **SNMP uses which port by default?**
   A. 161
   B. 25
   C. 80
   D. 110

5. **Which enumeration type identifies valid email users?**
   A. SMB Enumeration
   B. DNS Enumeration
   C. SNMP Enumeration
   D. SMTP Enumeration

6. **Which version of SNMP is considered most secure?**
   A. SNMPv1
   B. SNMPv2
   C. SNMPv3
   D. All versions are equal

7. **NetBIOS operates on which port?**
   A. 137–139
   B. 21–23
   C. 53
   D. 389

8. **Which tool can enumerate RPC services?**
   A. rpcclient
   B. Dirbuster
   C. John the Ripper
   D. Ntop

9. **DNS enumeration helps attackers identify:**
   A. Password hashes
   B. System files
   C. Hostnames and IP mappings
   D. Processes running

10. **Which protocol reveals device information such as uptime or system name?**
    A. SMTP
    B. SNMP
    C. FTP
    D. SSH

Topic 3 : Desktop & Server OS Vulnerabilities

1. **Which is a common OS vulnerability?**
   A. Strong encryption
   B. Patch management
   C. Buffer overflow
   D. Multi-factor authentication

2. **EternalBlue exploited which protocol?**
   A. FTP
   B. SMB
   C. Telnet
   D. RDP

3. **Which OS is affected by Dirty COW?**
   A. Windows
   B. Linux
   C. macOS
   D. Android only

4. **Privilege escalation occurs when:**
   A. Logs are deleted
   B. User gains higher privileges
   C. Password is reset
   D. Malware is removed

5. **Which tool is commonly used to exploit OS vulnerabilities?**
   A. Metasploit
   B. Wireshark
   C. Putty
   D. Packet Tracer

6. **Unpatched systems are vulnerable due to:**
   A. Updated firmware
   B. Missing security updates
   C. Strong group policies
   D. Antivirus installation

7. **Windows “PrintNightmare” vulnerability affects:**
   A. Task Scheduler
   B. Print Spooler service
   C. File Explorer
   D. Windows Defender

8. **Which is a mitigation technique?**
   A. Disabling firewall
   B. Using outdated systems
   C. Applying patches
   D. Allowing default passwords

9. **Authentication bypass allows attackers to:**
   A. Strengthen security
   B. Login without valid credentials
   C. Enhance encryption
   D. Improve OS stability

10. **DLL injection affects which type of system?**
    A. Non-executable systems
    B. Windows-based systems
    C. Firewalls
    D. Network switches

Topic 4 : Hacking Web Servers

1. **Which tool scans for web server vulnerabilities?**
   A. Nikto
   B. MS Paint
   C. Snort
   D. Nessus Agent only

2. **Which attack allows viewing arbitrary server files?**
   A. SQL Injection
   B. Local File Inclusion
   C. XSS
   D. Session Fixation

3. **Which server software is widely used?**
   A. Apache
   B. Notepad
   C. Chrome
   D. LibreOffice

4. **SQL Injection targets:**
   A. Operating system files
   B. Databases
   C. Network routers
   D. USB devices

5. **Directory traversal attack uses which sequence?**
   A. http://site.com/secure
   B. ../../etc/passwd
   C. SELECT * FROM users
   D. \

6. **Cross-Site Scripting (XSS) impacts:**
   A. Database servers
   B. User browsers
   C. VPN tunnels
   D. Firewalls

7. **RFI allows attackers to:**
   A. Upload documents
   B. Include remote malicious files
   C. Scan network ports
   D. Enumerate shares

8. **Which tool is commonly used for penetration testing web apps?**
   A. Burp Suite
   B. Wireshark
   C. Zenmap
   D. Cain & Abel

9. **A misconfigured server may leak:**
   A. Network diagrams
   B. Executable permissions
   C. Directory indexes
   D. Phishing data

10. **HTTPS provides:**
    A. File inclusion
    B. Encrypted communication
    C. Remote exploitation
    D. Password cracking

Topic 5 : Network Protection Systems

1. **A firewall operates mainly at which OSI layer?**
   A. Application
   B. Network
   C. Presentation
   D. Physical

2. **IDS stands for:**
   A. Internal Data Security
   B. Intrusion Detection System
   C. Internet Deployment System
   D. Internal Defense Software

3. **IPS differs from IDS because it can:**
   A. Detect only
   B. Block attacks
   C. Provide Wi-Fi access
   D. Store logs permanently

4. **A SIEM system is used for:**
   A. Log correlation
   B. Image editing
   C. Video compression
   D. File fragmentation

5. **Which is an example of a firewall?**
   A. Suricata
   B. pfSense
   C. Nessus
   D. John the Ripper

6. **EDR stands for:**
   A. Enterprise Data Relay
   B. Endpoint Detection & Response
   C. External Data Review
   D. Electronic Data Router

7. **Which security measure prevents unauthorized network access?**
   A. Screensaver
   B. NAC
   C. Defrag
   D. Zoom meetings

8. **NGFWs are known for:**
   A. Basic rule filtering
   B. Deep packet inspection
   C. File compression
   D. Password generation

9. **VPN provides:**
   A. Anonymity only
   B. Secure encrypted tunnels
   C. File storage
   D. Wireless access

10. **SIEM tools help detect attacks by:**
    A. Ignoring logs
    B. Correlating events
    C. Deleting audit trails
    D. Reducing encryption